

MASTER JAVA E-BOOK SERIES INDEX

జావా ఇంటర్వ్యూ + డెవలప్‌మెంట్ మాస్టర్ సిరీస్ (Beginner → 10+ Years Senior/Architect Java Full Stack Engineer)

Status:  **COMPLETE** — all 24 core books plus the special supplementary Book 15A have been fully generated, chapter by chapter, in Telugu + English, and are committed to this repository. **Language policy:** Telugu = primary teaching language. English = technical terms, code, APIs, interview answers (as specified). **Output format note:** This environment is a Linux git repository, not a Windows filesystem, so books are authored as clean, PDF-ready Markdown files inside this repo (folder structure below) instead of `F:\JAVA INTERVIEWS\...\pdf`. Each Markdown file is structured (headings, code blocks, tables) so it converts cleanly to PDF later (e.g., via Pandoc) — file naming numbers/order match the original spec, with `.md` instead of `.pdf`.

★ Recruiter-Priority Reweighting (applied from Book 15A onward)

Partway through this series, recruiter-search data for the target role — **Java Full Stack Developer, 3–5 years, 6–9 LPA** — showed Spring Boot, J2EE, Java, Java+Spring Boot, and Microservices as the most-searched keywords. From that point forward, every book was deliberately weighted toward this exact chain rather than treated as a generic Java curriculum:

```
Java foundation → Advanced Java / J2EE → ★ Spring Framework → ★★★★★ Spring Boot
→ ★★★★★ Spring Boot + REST + JPA/Hibernate + Security → ★★★★★ Microservices
→ Kafka + Redis + Docker + Kubernetes + Cloud + CI/CD → ★ Java Full Stack
→ Real-world projects + production architecture + interviews
```

This is why **Book 15A** exists as a standalone special addition (Full Stack frontend/integration — HTML/CSS/JS/TypeScript/React connected to the Spring Boot backend), why **Books 16–17** give Spring Cloud Gateway/Eureka/Config Server/Resilience4j and Kafka full dedicated depth rather than a single chapter each, and why **Books 18–24** consistently trace every pattern, algorithm, and design decision back into this Spring Boot/Microservices core rather than treating it as separate material.

Folder / File Structure (as actually generated)

```
java-interview-ebook-series/  
  00_MASTER_INDEX.md                                <- this file  
  01_Java_Fundamentals_Telugu_English.md  
  02_Java_OOPS_Mastery_Telugu_English.md  
  03_JVM_Memory_Management.md  
  04_Java_Exception_Handling.md  
  05_Java_Collections_Framework.md  
  06_Java_Generics.md  
  07_Java_8_Plus_Modern_Java.md  
  08_Java_Multithreading_Concurrency_Part_1.md  
  08_Java_Multithreading_Concurrency_Part_2.md  
  09_JDBC_Database.md  
  10_Advanced_Java_Backend.md  
  11_Spring_Core.md  
  12_Spring_Boot.md  
  13_Spring_Data_JPA_Hibernate.md  
  14_Spring_Security.md  
  15_Java_Testing.md  
  15A_Java_Full_Stack_Frontend_Integration.md        <- ★ SPECIAL supplementary book  
(recruiter-priority addition)  
  16_Microservices.md  
  17_Kafka_Event_Driven_Architecture.md  
  18_Java_Design_Patterns.md  
  19_Low_Level_Design.md  
  20_DSA_Pattern_Mastery.md  
  21_System_Design_HLD.md  
  22_System_Design_Case_Studies.md  
  23_Java_Interview_Master_Book.md  
  24_Production_Java_Project.md
```

25 files total: 24 core books + 1 special supplementary book.

Series-Wide Standards (applies to every book)

Every book includes: Title page · TOC · Learning objectives · Prerequisites · Telugu + English dual explanation per concept · Runnable Java code + output · Internal working (JVM/memory/framework internals where relevant) · Diagrams (ASCII) · Real-world/production examples · Interview answers (with Deep-Senior extensions where warranted) · Cross-questions · Tricky questions · 6-level exercises (Beginner→Mastery) · Cheat sheet · One-Day & One-Week revision plans · Final mastery checklist.

Java versions are always explicitly labeled: **Java 7** (not LTS), **Java 8**, **Java 11 (LTS)**, **Java 17 (LTS)**, **Java 21 (LTS)** — features tagged by version, never conflated.

Every book, from Book 12 onward, also explicitly cross-references earlier books by number and chapter — the series is designed as one continuous, cumulative mental model rather than 25 standalone documents.



MASTER BOOK TABLE

#	Book Name	Prerequisites	Core Topics (summary)	Java Versions	Chapters	Difficulty	Practical Projects	Interview Level
01	Java Fundamentals — Beginner to Professional	None	Syntax, variables, data types, operators, control flow, arrays, methods, classes/objects basics, String family, wrapper classes, enums, nested/inner/anonymous classes	7/8	16	Beginner	Student Management Console App	Fresher
02	Java OOPS — Beginner to 10+ Yrs Interview Mastery	Book 01	Encapsulation, abstraction, inheritance, polymorphism, overloading/overriding, abstract class vs interface, constructors/chaining, composition/aggregation/association, SOLID, coupling/cohesion	7/8	14	Beginner → Intermediate	Library Management System	Fresher–Mid
03	JVM & Java Memory Management — Internals Mastery	Book 01	JDK/JRE/JVM, classloading, bytecode, execution engine, JIT, heap/stack/metaspaces/PC register, GC algorithms, OOM/StackOverflow, profiling	8/11/21	10	Intermediate	Heap dump/profiling case study	Mid–Senior
04	Java Exception Handling	Books 01–02	Exception hierarchy, checked/unchecked, try/catch/finally, throw/throws, custom exceptions, multi-catch, try-with-resources, propagation, production error handling	7/8/11	8	Beginner → Intermediate	Order Processing System	Fresher–Mid
05	Java Collections Framework	Books 01–04	List/Set/Map/Queue family, hashing internals, equals/hashCode, resizing/load factor/treeification, Comparable/Comparator, Big-O decision framework	8/11	14	Intermediate	In-Memory Inventory System	Mid–Senior
06	Java Generics	Books 01–05	Generic classes/methods, wildcards, bounded types, PECS, type erasure	8	6	Intermediate	Generic In-Memory Data Access Layer	Mid
07	Java 8+ Modern Java Mastery	Books 01–06	Lambdas, functional interfaces, Streams, Optional, method references, Date/Time API, CompletableFuture; Java 11/17/21 highlights (records, sealed types, pattern matching, text blocks, virtual threads)	8 (deep), 11, 17, 21	16	Intermediate → Advanced	Order Analytics Pipeline	Mid–Senior
08	Java Multithreading & Concurrency (Parts 1 & 2)	Books 01–07	Thread lifecycle, Runnable/Callable/Future, ExecutorService, synchronized/volatile/atomics, locks, CountdownLatch/CyclicBarrier/Semaphore, BlockingQueue, JMM/happens-before, deadlock/livelock/starvation, CompletableFuture, Virtual Threads	8/11/21	20 (2 parts)	Advanced	Concurrent Order Processing System	Senior
09	JDBC & Database	Books 01–06	JDBC architecture, Connection/Statement/PreparedStatement, ResultSet, transactions/ACID, connection pooling, SQL injection, indexes/joins/normalization, isolation levels, DAO pattern	8/11	10	Intermediate	Banking Ledger CRUD Application	Mid

#	Book Name	Prerequisites	Core Topics (summary)	Java Versions	Chapters	Difficulty	Practical Projects	Interview Level
10	Advanced Java / Backend Fundamentals	Books 01–09	HTTP, Servlets, sessions/cookies, filters/listeners, REST basics, JSON, JWT structure, RBAC, layered architecture	8/11	10	Intermediate→Advanced	Servlet-Based Task Manager	Mid–Senior
11	Spring Core	Books 01–10	IoC, DI, Bean lifecycle/scopes, ApplicationContext, annotations, component scanning, AOP, circular dependency resolution	8/11/17+	10	Intermediate	Plain Spring DI Application	Mid
12	Spring Boot Production Mastery	Book 11	Auto-configuration, starters, REST controllers, DTOs, Bean Validation, global exception handling, Actuator, packaging/Docker	17/21	14	Advanced	Production-Style REST API	Mid–Senior
13	Spring Data JPA / Hibernate	Book 12	Entity/persistence context, relationships (1:1,1:N,N:1,N:N), lazy/eager, N+1, cascading, @Transactional/AOP mechanics, JPQL, pagination, performance	17/21	12	Advanced	E-commerce Data Model	Senior
14	Spring Security	Books 12–13	Filter chain, UserDetailsService, password hashing, JWT stateless auth, refresh tokens, OAuth2, CORS, CSRF, production security checklist	17/21	10	Advanced	JWT-Secured REST API	Senior
15	Java Testing	Books 12–14	Test pyramid, JUnit 5, Mockito, AAA/FIRST, Spring Boot test slices (@WebMvcTest/@DataJpaTest), Testcontainers, risk-based test strategy	8/11/17+	8	Intermediate→Advanced	Full Test Suite (Books 12–14)	Mid–Senior
15	★ SPECIAL: Java Full Stack Developer — Frontend & Integration	Books 01–14	HTML5, CSS3/responsive design, modern JavaScript (ES6+), TypeScript, React (components/hooks), connecting React to Spring Boot REST APIs, JWT auth end-to-end, state management, full-stack deployment architecture	Frontend: modern JS/TS/React; Backend: as above	11	Advanced	Full Stack Task Manager (Books 12–14 backend + React frontend)	Mid–Senior
16	Microservices	Books 12–15	Monolith vs microservices, bounded contexts, sync/async comms, ★★★★★ Config Server, ★★★★★ Eureka service discovery, ★★★★★ Spring Cloud Gateway, ★★★★★ Resilience4j (Circuit Breaker/Retry/Bulkhead/Timeout), SAGA, CQRS, Event Sourcing, tracing/logging, Docker, Kubernetes fundamentals	17/21	14	Advanced	Order+Payment+Notification Microservices System	Senior
17	Kafka & Event-Driven Architecture	Book 16	Broker/topic/partition, producer/consumer/consumer group, offsets, replication/ISR, delivery semantics (at-most/at-least/exactly-once), idempotency, retry/DLQ, Spring Kafka	Kafka + Java 17/21	10	Advanced	Event-Driven Order→Payment→Notification Pipeline	Senior

#	Book Name	Prerequisites	Core Topics (summary)	Java Versions	Chapters	Difficulty	Practical Projects	Interview Level
18	Java Design Patterns	Books 01–07, 11–17	Creational (Singleton, Factory, Abstract Factory, Builder, Prototype), Structural (Adapter, Decorator, Facade, Proxy, Composite), Behavioral (Observer, Strategy, Template Method, Command, State, Chain of Responsibility) — every pattern traced into Spring/Spring Boot/Microservices/Kafka internals	8/17/21	16	Intermediate→Advanced	Refactor a bad-design mini app (OrderProcessingSystem) pattern by pattern	Mid–Senior
19	Low Level Design	Book 18	SOLID/UML recap and a 7-step LLD approach; full LLD for Parking Lot, BookMyShow, Tic-Tac-Toe, Elevator, ATM, Splitwise, Library, Food Delivery, Payment System	Language-agnostic + Java	10 (case-study driven)	Advanced	9 complete LLD case studies	Senior
20	DSA Pattern Mastery	Book 01, Book 06	Complexity analysis, prefix sum, two pointers, sliding window, sorting, binary search, hashing, string patterns, linked list, monotonic stack/queue, recursion, backtracking, greedy, DP (1D/2D/Knapsack), closing pattern-recognition decision framework	Java 8+	20 (by pattern)	Beginner→Advanced	Curated problem sets per pattern	Fresher–Senior
21	System Design / HLD	Books 16–17, 19	7-step HLD approach, scaling/load balancing, capacity estimation, SQL/NoSQL, CAP theorem, replication/sharding, caching/Redis, CDN, async messaging, reliability at scale, microservices patterns generalized	Concept-level	12	Advanced	Design exercises per section + full URL-shortener walkthrough	Senior–Architect
22	System Design Case Studies	Book 21	Full HLD for WhatsApp, Netflix, Hotstar, Instagram, Rate Limiter — requirements → estimation → API/DB design → caching/scaling/messaging → trade-offs	Concept-level	5 (one per system)	Advanced	5 complete system designs	Senior–Architect
23	Java Interview Master Book	All prior books	Beginner→Architect question bank organized by experience level (Fresher/Junior/Mid-Level/Senior/Architect): direct/why/how/difference/scenario/coding/debugging/design/cross/trick questions, each with short + professional + deep-senior answers	All	Organized by level, not chapters	All levels	Mock-interview drills	Fresher–Architect
24	Production Java Project	All prior books + 15A	ShopSphere: Java 21 + Spring Boot + Spring Data JPA + PostgreSQL + Redis + Kafka + Spring Security/JWT + Docker + testing + logging/monitoring — full source walkthrough, 12 production modules, production-readiness checklist	21	12 (by module)	Advanced	1 full production-grade backend system (capstone)	Senior

Dependency Chain (build order, as actually generated)

```
01 Fundamentals
└─ 02 OOPS
    └─ 03 JVM/Memory    04 Exceptions
        └─ 05 Collections → 06 Generics → 07 Java 8+
            └─ 08 Concurrency (Parts 1–2)
                └─ 09 JDBC → 10 Advanced Java/Backend
                    └─ 11 Spring Core → 12 Spring Boot
                        └─ 13 JPA/Hibernate → 14 Spring Security → 15 Testing
                            └─ 15A ★ Full Stack Frontend & Integration

(special)
                                └─ 16 Microservices → 17 Kafka
                                    └─ 18 Design Patterns → 19 LLD
                                        └─ 20 DSA (studied in parallel)

from Book 01/06)
                                                └─ 21 System Design/HLD →

22 Case Studies
                                                    └─ 23 Interview

Master Book
                                                        └─ 24

Production Project (capstone)
```

DSA (Book 20) only hard-requires Book 01 + Book 06, so it can be studied in parallel with the Core Java / Spring track rather than strictly after Book 19.

Series Completion Summary

This series totals **~340 chapters/modules across 25 files** (24 core books + Book 15A), fully generated in Telugu + English with runnable code, diagrams, interview Q&A, exercises, and revision plans throughout. It was built and approved book-by-book, with a deliberate mid-series recruiter-priority reweighting (see above) applied consistently from Book 15A through the Book 24 capstone.

What the series delivers, start to finish:

- **Books 01–10:** Core Java → JVM internals → Exceptions → Collections → Generics → Java 8+ → Concurrency → JDBC → Backend fundamentals
- **Books 11–15 + 15A:** Spring Core → Spring Boot → JPA/Hibernate → Spring Security → Testing → ★ Full Stack Frontend & Integration (special)
- **Books 16–17:** Microservices (Gateway/Discovery/Config/Resilience) → Kafka & Event-Driven Architecture
- **Books 18–20:** Design Patterns → Low-Level Design → DSA Pattern Mastery
- **Books 21–22:** System Design/HLD → 5 complete real-world case studies
- **Books 23–24:** Fresher → Architect Interview Master Book → ShopSphere production capstone project

The Book 24 capstone (`24_Production_Java_Project.md`) closes with a full source map crediting every one of Books 01–23 and 15A for its specific, traceable contribution to the finished production system — the intended proof that this is one continuous body of knowledge, not 25 disconnected documents.